

THOMAS DAGÈS

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EXPERIENCE

Technical University Munich – Munich Centre of Machine Learning

2024 — Today

Postdoc – Thomas Bayes Fellow — Geometric Computer Vision

Munich, Germany

Host: Prof. Daniel Cremers

Technion, Faculty of Computer Science

2024 – 2025

Postdoc — Geometric Computer Vision

Haifa, Israel

Hosts: Profs. Ron Kimmel, Alfred M. Bruckstein, Michael Lindenbaum

EDUCATION

Technion, Faculty of Computer Science

2020 – 2024

PhD — Geo-Metric Learning and the Value of Models

Haifa, Israel

Advisors: Profs. Alfred M. Bruckstein and Michael Lindenbaum

Technion, Faculty of Computer Science

2017 – 2019

MSc — Spatio-Temporal Salient Detection of Moving Objects in Highly Degraded Video Data

Haifa, Israel

Advisors: Profs. Alfred M. Bruckstein and Michael Lindenbaum

École Polytechnique

2014 – 2017

Engineering degree (MSc equivalent) specialised in Mathematics and Computer Science

Palaiseau, France

Lycée Louis-Le-Grand

2012 – 2014

Intensive preparatory classes for competitive entrance examinations to the French Grandes Écoles

PUBLICATIONS

- **Thomas Dagès**, Michael Lindenbaum, and Alfred M. Bruckstein. “Metric Convolutions: A Unifying Theory to Adaptive Image Convolutions”, *ICCV* (2025).
- **Thomas Dagès***, Simon Weber*, Ya-Wei Eileen Lin, Ronen Talmon, Daniel Cremers, Michael Lindenbaum, Alfred M. Bruckstein, and Ron Kimmel. “Finsler Multi-Dimensional Scaling: Manifold Learning for Asymmetric Dimensionality Reduction and Embedding”, *CVPR* (2025).
- Amit Bracha*, **Thomas Dagès***, and Ron Kimmel. “Wormhole Loss for Partial Shape Matching”, *NeurIPS* (2024).
- Amit Bracha, **Thomas Dagès**, and Ron Kimmel. “On Unsupervised Partial Shape Correspondence”, *ACCV* (2024).
- **Thomas Dagès**, Laurent D. Cohen, and Alfred M. Bruckstein. “A model is worth tens of thousands of examples for estimation and thousands for classification.” *Pattern Recognition journal* (2024).
- Simon Weber*, **Thomas Dagès***, Maolin Gao, and Daniel Cremers. “Finsler Laplace-Beltrami Operator for Shape Correspondence”, *CVPR* (2024).
- **Thomas Dagès**, Michael Lindenbaum, and Alfred M. Bruckstein. “From Compass and Ruler to Convolution and Nonlinearity: On the Surprising Difficulty of Understanding a Simple CNN Solving a Simple Geometric Estimation Task”, *arXiv preprint arXiv:2303.06638* (2023), *under review*.

- **Thomas Dagès**, Laurent D. Cohen, and Alfred M. Bruckstein. “A Model is Worth Tens of Thousands of Examples.” *SSVM* (2023) .
- **Thomas Dagès** and Alfred M. Bruckstein. “Doubly Stochastic Pairwise Interactions for Agreement and Alignment.” *SIAM Journal on Applied Mathematics* 82.4 (2022): 1246-1266.
- Ariel Barel, **Thomas Dagès**, Rotem Manor, and Alfred M. Bruckstein. “Probabilistic gathering of agents with simple sensors.” *SIAM Journal on Applied Mathematics* 81.2 (2021): 620-640.
- **Thomas Dagès** and Alfred M. Bruckstein. “A Bound on the Edge-Flipping Distance between Triangulations (Revisiting the Proof).” *arXiv preprint arXiv:2106.14408* (2021).
- **Thomas Dagès**, Michael Lindenbaum, and Alfred M. Bruckstein. “Seeing Things in Random-Dot Videos.” *MSc Thesis. Computer Science Department, Technion* (2020).
- **Thomas Dagès**, Michael Lindenbaum, and Alfred M. Bruckstein. “Seeing Things in Random-Dot Videos.” *Asian Conference on Pattern Recognition. Springer, Cham* (2019).

SUPERVISION EXPERIENCE

Pointcloud Laplacians	01/01/2025 – Today
• MSc and PhD project.	
Revisiting Multi-Dimensional Scaling	01/10/2024 – Today
• MSc project.	
Automated Saffron Detection and Extraction for a Factory Line	01/10/2021 – Today
• Undergraduate and MSc project.	

TEACHING EXPERIENCE

236201 - Introduction to Data Processing and Representation	2025 – Today
• Lecturer.	
• Giving the lectures and fully managing teaching duties for the course.	
236201 - Introduction to Data Processing and Representation	2021 – 2025
• Head Teaching Assistant.	
• Creating and teaching entirely new material, homeworks, and exams and correcting the latter.	
236200 - Signal, Image, and Data Processing	2019 – 2021
• Cohead Teaching Assistant.	
• Teaching with mostly existing material, creating and correcting new homeworks and exams.	
236861 - Geometric Computer Vision	2019
• Homework Checker.	

RESEARCH SIDE PROJECTS

Technion	2018
<i>Student Research Project</i>	<i>Haifa, Israel</i>
• Created a state-of-the-art astronomical image deblurring algorithm by applying an approach using sparse representations.	
École Polytechnique – Université Paris Descartes	2016 – 2017
<i>Student Research Project</i>	<i>Palaiseau, France</i>
• Created and implemented an emotion detector on human faces with a webcam by using the active shape model and fuzzy logic.	
Dassault systèmes	2017
<i>Research Intern</i>	<i>Vélizy, France</i>

- In the context of 3D city modelling based on ground photographs, created and implemented an occlusion and reflection remover on the reconstructed texture with automatic detection and inpainting.

Renault Technologies Romania

2016

Computer-Aided Engineering Intern

Bucharest, Romania

- Numerically modelled and simulated the problems of exaggerated opening and closing of a car bonnet maintained by a prop rod, for replacing the costly hydraulic cylinders on the Renault Kaptur.

RESEARCH INTERESTS

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| • Computer Vision | • Modelling Human Vision | • Interpretable Artificial Intelligence | • Planar and Differential Geometry |
| • Signal and Image Processing | • Classical Machine Learning | • Abstract and Applied Mathematics | • Statistics |
| • Image Analysis | | | • Applied Computer Science |

LANGUAGES

Computer: Python, Matlab, C++, Java, OCaml

Human: English (bilingual), French (bilingual), German (intermediate), Russian (beginner), Hebrew (beginner)

EXTRACURRICULAR

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| • Sports, often with a group: horse riding, bouldering, surfing. | • Soothing manual activities like baking, calligraphy, and letter sealing. |
| • Thought provoking Sci-fi reading/viewing and discussions. | • Unplugging through gaming and outings with friends. |